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Ethernet, X-Surf and PanoTCP on the G2 Amiga

The G2's Ethernet port carries the firmware's network console and time service, an X-Surf 100 board for AmiTCP or Roadshow, and PanoTCP: a bsdsocket.library whose TCP/IP stack runs on the RISC-V firmware.

HIGHLIGHTS

- Network console on TCP 2323, discovery beacon, SNTP battery clock.
- X-Surf 100 (NE2000) board for Amiga TCP/IP stacks.
- PanoTCP: bsdsocket.library 4 on the firmware's lwIP, about 350 KB/s.

KEY PARAMETERS

PARAMETER	VALUE
Link	10/100 Mbit/s
Console	TCP 2323
Beacon	UDP 48951
X-Surf rx / tx	124 / 67 KB/s
PanoTCP rx / tx	329 / 375 KB/s

1 ETHERNET AND THE FIRMWARE

ITEM	VALUE
PHY, MAC	Marvell 88E1119R; the fabric's Ethernet MAC at 10/100 Mbit/s
Stack	lwIP 2.2.0 (BSD-3-Clause) in the firmware: DHCP, TCP, UDP, DNS
Network console	TCP port 2323: the firmware's commands (status, log, ls, get, put, rm, reset, reboot, golden, afs, df, adf, key, snap, vid, midi, ftdi, date, sock, xsurf)
Discovery	A UDP broadcast every second on port 48951: PANO mac=… ip=… port=2323 build=… board=g2
Time	SNTP at start-up: the battery clock ($DC0000, MSM6242) loaded with network time plus TZ=; NTPWAIT= delays the first start up to 120 s

2 X-SURF 100 BOARD (AMITCP, ROADSHOW)

ITEM	VALUE
Board	An NE2000 (RTL8019AS) model as the Zorro II X-Surf 100; driver x-surf-100.device
Frames	Relayed by the firmware between the board's host mailbox and the MAC; 16-slot receive queue
Station address	The unit's factory MAC with the locally administered bit set
Measured	AmiTCP 3.0b2: ping from the LAN 0 % loss, 74–134 ms; 1 MB received at 124 KB/s, sent at 67 KB/s

3 PANOTCP: BSDSOCKET.LIBRARY ON THE FIRMWARE

ITEM	VALUE
Function	C:PanoTCP installs bsdsocket.library version 4; each call is an operation of the firmware's socket service on lwIP. The 68020 computes no checksum and keeps no timer
Transport	Blocks 511–895 of HD0 through scsi.device (the socket window), never written to the image
Programs run	LumiFTP 1.2 (FTP), bebboscp 1.46.1 and bebbossh (SFTP, SSH), name lookup
Measured	Receive 315–329 KB/s, send 355–375 KB/s (2 MB, LAN); LumiFTP 2,800,640 bytes down in 14.9 s (183 KB/s), up 226 KB/s, both equal to the server's file; SFTP 32 KB/s (the 68020's decryption); DNS 18–63 ms
Limits	Sector port about 0.78 ms a block (650 KB/s); TCP window toward the unit 8 segments

4 LIMITS AND UNTESTED ITEMS

ITEM	STATE
Gigabit link	Not used: 10/100 advertised
PanoTCP listening sockets, UDP sendto/recvfrom, ObtainSocket, getaddrinfo	Written in the firmware; untested from AmigaOS
PanoTCP started from S:User-Startup	Installed by a start script on the test unit; the measurements above started it by hand
X-Surf register alias at 0x800 (Linux, NetBSD drivers)	Not decoded

Measured on a G2 revision C unit with firmware release 1.0 and fabric 0x0AB5EF2D unless stated. Repository mikewolak/pano-g2-amiga, release 1.0.